



Hydrodynamic analysis of nanofluids flow over 45° inclined porous square cylinder using Darcy–Brinkman–Forschheimer model

Jaspinder Kaur¹ | Jatinder Kumar Ratan¹ | Roderick Melnik² | Anurag Kumar Tiwari¹

¹Department of Chemical Engineering,
Dr. B.R. Ambedkar National Institute of
Technology, Jalandhar, India

²MS2Discovery Interdisciplinary Research
Institute, Wilfrid Laurier University,
Waterloo, Canada

Correspondence

Anurag Kumar Tiwari, Department of
Chemical Engineering,
Dr. B.R. Ambedkar National Institute of
Technology, Jalandhar 144011, Punjab,
India.
Email: tiwaria@nitj.ac.in

Abstract

Various engineering applications commonly involve the flow of nanofluids over a porous 45° inclined square cylinder. Therefore, the current study is to assess the impact of the Darcy parameter (Da), nanoparticle volume fraction (ϕ), and Reynolds number (Re) on the momentum transport characteristics over the 45° inclined porous square cylinder. The governing equations were solved numerically using the Darcy–Brinkman–Forschheimer model for different values of Da ($10^{-6} \leq Da \leq 10^{-2}$), nanoparticle volume fraction ϕ ($0.01 \leq \phi \leq 0.05$), and Reynolds number Re ($1 \leq Re \leq 40$). The results for each parameter were visualized using streamline plots, velocity profiles within the porous cylinder, and vorticity contours. Complex flow behaviors were observed between $Da = 10^{-3}$ – 10^{-2} , flow separation detached and disappeared at the cylinder's downstream side at critical Darcy number. Drag and pressure coefficients are used to represent the global and local parameter of the flow fields. The pressure coefficient on the surface of the cylinder showed an inverse relationship with the nanoparticles volume fraction and Darcy number. The strength of the drag coefficient decreased with the addition of nanoparticles to the base fluid, with a decreasing trend observed for $Da = 10^{-4}$ – 10^{-2} and no change observed for $Da = 10^{-6}$ – 10^{-4} . At last, the comparative analysis has been conducted between a porous square cylinder at 0° inclined and another 45° inclined.

KEYWORDS

45° inclined porous square, Darcy number, drag coefficient, nanofluids

1 | INTRODUCTION

Momentum transport over the porous media is found in various fields of science and engineering. Porous media play a vital role in various fields, including heat and mass transfer, filtration, and oil and gas production.¹ Therefore, a fundamental understanding of transport behaviors

across and through the porous media is very important. The porous media refers to a type of material that has a high internal surface area due to the presence of pores or voids. This material can be either solid or permeable and can be found in various forms, such as foams, sintered metals, and natural materials like rock formations.² The study of fluid flow over and through porous media is



known as porous flow and is an important area of research in fluid dynamics. The approach to solving flow over porous media involves the development of mathematical models to recognize the complex interactions between the fluid and the porous material. These models and methods consider factors such as the porosity of the medium, the fluid viscosity, and the permeability of the pores. The various ranges of porous media are available to explore, specifically the porous cylinders of different cross-sections. When a fluid flow interacts with a porous cylinder, complicated flow patterns develop nearby the cylinder. Flow over a porous cylinder provides important insights into fluid dynamics with applications in a variety of fields, from engineering to environmental science.³ Therefore, a variety of applications excites the researcher to explore the reason behind that. Due to its practical importance, a considerable amount of research has been dedicated to investigating fluid flow over a porous cylinder with different cross-sectional shapes such as a sphere, cylinder, square, and so on. In general, two models are used in the modeling of fluid flow over the porous cylinder such as one-domain approach and two-domain approaches depending upon the problem and situation.

In recent years, several research studies have examined the hydrodynamic analysis of flow over a porous cylinder, particularly a porous circular or square cylinder, at various Re . Only some of the works are cited here. Bhat-tacharya et al.⁴ have examined the flow over and through a porous cylinder. They found that there is reduction in the drag coefficient (C_D). As they increased the value of Re , the value of drag coefficient decreases, while decreasing the value of Da resulted in a rise in the drag coefficient. A study on the wake structure behind a porous circular cylinder performed by.⁵ Their findings revealed that the size of the recirculation reduced as Da increased, and for certain values of Da , the recirculating wakes initially emerged within the porous medium.

Other than the porous circular cylinders, several studies have investigated for the flow behaviour around the porous square cylinders. For example, Chen et al.⁶ examined the flow around a porous square cylinder using stress-jump boundary conditions to model the interface between the fluid and porous layers. They found that the drag coefficient decreases with an increase in porosity. A numerical analysis was conducted by Dhinakaran and Ponmozhi⁷ to investigate the heat transfer and fluid flow phenomena in a porous square cylinder. Their findings indicated that, for a Darcy number of approximately 10^{-6} , the drag coefficient (C_D) and wake length (L_r) of this particular cylinder material approached those of a solid square cylinder.⁸ conducted an investigated in to the MHD flow and heat transfer around a solid cylinder

featuring a porous ring. To calculate the average Nusselt number, the least square method were utilized and two empirical equations were developed, which accounted for the effects of the magnetic field and Darcy numbers. A mesoscopic approach was employed by Vijaybabu et al.⁹ to examine the flow and mixed convection heat transfer from a porous square cylinder under aiding buoyancy. The analysis involved a range of Reynolds numbers (Re) and Darcy numbers (Da) from 2 to 40 and 10^{-6} to 10^{-2} , respectively. The study revealed that as the permeability levels increase, the L_r and C_D values monotonically decrease. Furthermore, aiding buoyancy leads to a significant reduction in L_r and an increase in C_D values.

In addition, there is a set of studies focused on investigating the significant features of nanofluid flow over a porous cylinder. With the addition of nanoparticles in base fluids, an interesting behavior has been observed due to enhanced thermo-physical properties.^{10,11} The nanofluids flow through porous media is a complex phenomenon and use in various industrial and technological applications.^{1,12} This type of flow is characterized by the interaction between the nanofluids and the porous structure of the cylinder, which can have an important effect on the fluid's transport behavior. To effectively utilize nanofluids, it is important to understand the nanofluids flow behavior through porous cylinders, including the effect of fluid properties, porous structure, and operating conditions on the flow behavior. The addition of nanoparticles to a base fluid can affect the nanofluids flow in a porous media in several ways as the presence of nanoparticles can increase the viscosity of the fluid, which can make it more difficult for the fluid to flow through the porous medium. This increased viscosity can affect the porous media permeability and the rate of fluid flow. The size and concentration of the nanoparticles can also affect the nanofluids flow in porous media. If the nanoparticles are too large, they may deposit and clog the pores in the porous media, reducing the permeability and affecting the fluid flow behaviors. The interactions between the nanoparticles can also affect the nanofluids flow in porous media. For example, aggregation of the nanoparticles can increase the fluid viscosity and alter the fluid flow behavior.¹³ In summary, nanofluids can significantly affect the flow field near and through the porous media. It is important to consider these effects when designing and optimizing systems that involve the nanofluids flow in porous media.

In conclusion, the research carried out on the flow of nanofluids flow over a porous cylinder have revealed many exciting possibilities for improved flow analysis and thermal enhancement. Also very limited information is available related to non-circular cylinder specially cylinders of square cross section case which take the effect



of fixed separation point (such as sharp corner of the cylinder) and despite the simple geometrical shape, the flow over a square cylinder involves complex transport phenomena which directly impact the flow characteristics. On reviewing the literature, it was found that there are no specific studies that investigated the flow of nanofluids in 45° inclined porous square cylinders under different ranges of Reynolds numbers, nanoparticles volume fraction, and Darcy number. Hence, the primary focus of the current study is to investigate the effect of the Darcy parameter (Da), nanoparticles volume fraction (ϕ), and Reynolds number (Re) on the flow characteristics. Numerical results are investigated in a steady flow regime for different values of Da ($10^{-2} \leq Da \leq 10^{-6}$), nanoparticles volume fraction ($0.01 \leq \phi \leq 0.05$), and Reynolds number ($1 \leq Re \leq 40$) for fixed porosity ($\varepsilon = 0.9$) of the cylinder.

2 | PROBLEM FORMULATION AND MATHEMATICAL MODELING

The schematic diagram of the present numerical problem is depicted in Figure 1. Here, we considered a porous square cylinder (infinite along z -direction) inclined at 45° (B considered as side length) subjected in the uniform flow (from left to right) of nanofluids (CuO-Water) with velocity (V_∞) and temperature (T_∞). This infinite domain problem converted into an imaginary finite domain by considering the imaginary circular domain, which must be sufficiently far away from the cylinder. Uniformly dispersed CuO nanoparticles (spherical shape) with an average particle size of 30 nm are used in this numerical simulation. For the present numerical analysis, the single domain approach¹⁴ is adopted for both nanofluids and porous media. The physical properties of CuO

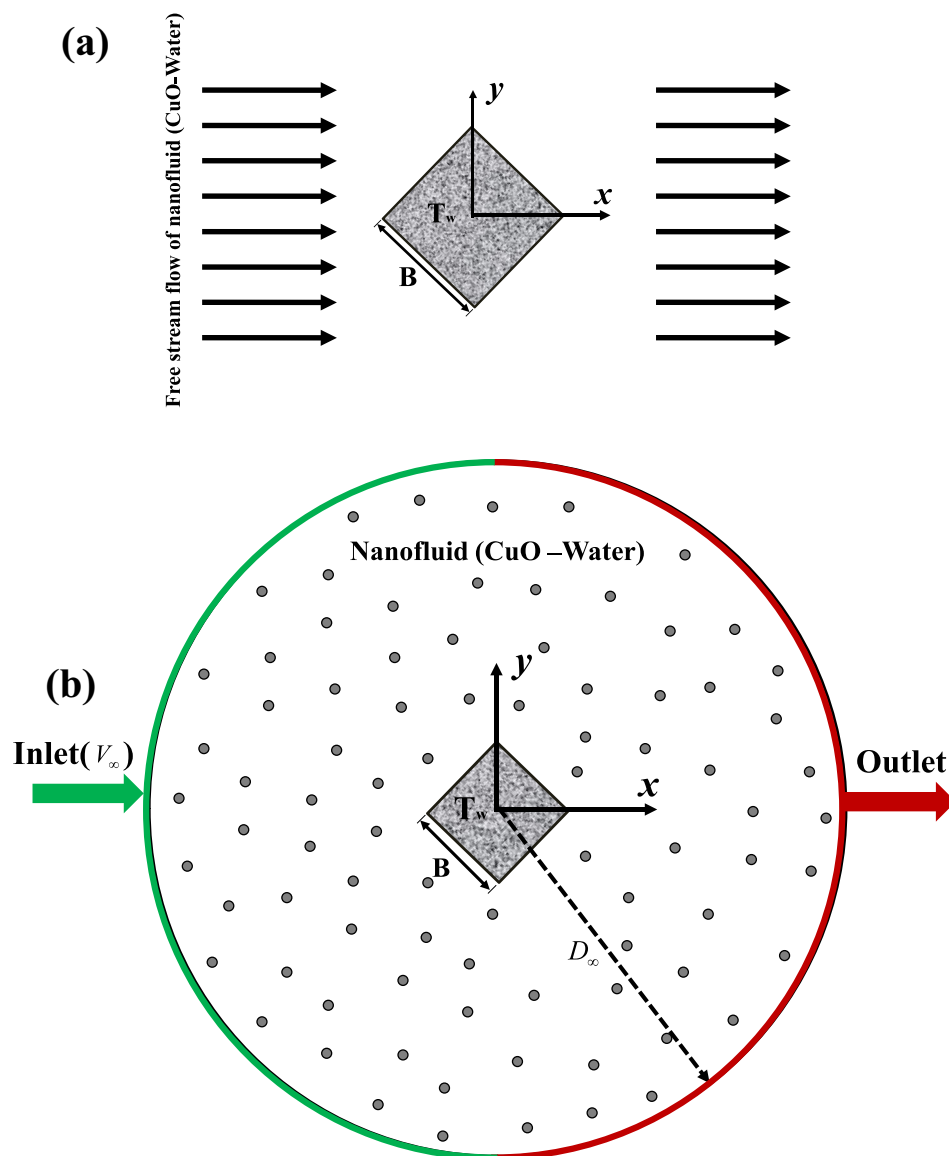


FIGURE 1 Schematic diagram of (a) flow nanofluid (CuO-water) over the 45° inclined porous square cylinder, (b) imaginary computational domain.



nanoparticles and water at the reference temperature are taken from literature.¹⁵ Under the consideration of 2D, laminar, steady flow of an incompressible nanofluids flow around and through the square cylinder of porous material has been written in a single set of equations for both porous media and fluid.

The Darcy–Brinkman–Forchheimer model¹⁶ is incorporated in the equation to capture the viscous and inertial effects in porous media. For porous cylinder, the following assumptions were consider for current numerical problem to make it numerical feasible.

- Porous cylinder is homogeneous and isotropic.
- Uniform permeability and porosity in 45° inclined porous square cylinder.
- Constant fluid property and negligible body force term.

The dimensionless governing equations are given as follows^{1,17}:

Continuity Equation:

$$\frac{\partial U}{\partial X} + \frac{\partial V}{\partial Y} = 0 \quad (1)$$

Momentum equation:

$$\frac{1}{\varepsilon^2} \left(U \frac{\partial U}{\partial X} + V \frac{\partial U}{\partial Y} \right) = -\frac{\partial P}{\partial X} + \frac{\Lambda}{\varepsilon \text{Re}} \left(\frac{\partial^2 U}{\partial X^2} + \frac{\partial^2 U}{\partial Y^2} \right) - \frac{1}{\text{ReDa}} U - \frac{F}{\sqrt{\text{Da}}} \left(\sqrt{U^2 + V^2} \right) U \quad (2)$$

$$\frac{1}{\varepsilon^2} \left(U \frac{\partial V}{\partial X} + V \frac{\partial V}{\partial Y} \right) = -\frac{\partial P}{\partial Y} + \frac{\Lambda}{\varepsilon \text{Re}} \left(\frac{\partial^2 V}{\partial X^2} + \frac{\partial^2 V}{\partial Y^2} \right) - \frac{1}{\text{ReDa}} V - \frac{F}{\sqrt{\text{Da}}} \left(\sqrt{U^2 + V^2} \right) V \quad (3)$$

Where $F = \frac{1.75}{\sqrt{150}} \frac{1}{\varepsilon^{2/3}}$ is the inertial factor.¹⁸

The following dimensional variables are introduced to non-dimensional the above governing equations (1)–(4):

$$X = \frac{x}{B}, Y = \frac{y}{B}, P = \frac{p}{\rho_{nf} V_\infty^2}, U = \frac{V_x}{V_\infty}, V = \frac{V_y}{V_\infty}, \Lambda = \mu_e / \mu_{nf}$$

The porosity ε of the cylinder is determined as the volume of vacant space in the porous cylinder for fluid flow¹⁹ and defined as,

$$\varepsilon = \begin{cases} 1, & \text{outside the porous object} \\ 0 < \varepsilon < 1, & \text{inside the porous object} \end{cases}$$

The Carman–Kozeny relation²⁰ is used to connect the porosity and permeability terms for porous material. The physical characteristics of nanofluids, including density (ρ_{nf}) and viscosity (μ_{nf}), are computed based on the nanoparticles volume concentration (ϕ), as well as the base fluid properties and the size of the nanoparticles. These properties are utilized to obtain numerical solutions in this investigation. To approximate the functional dependence as follows,

- The density of the nanofluids,²¹ $\rho_{nf} = (1 - \phi)\rho_{bf} + \phi\rho_{np}$
 - The viscosity of the nanofluids²²;

$$\mu_{nf} = \mu_{bf} + \frac{\rho_{np} \nu_B d_{np}^2}{72 B \chi} \quad (4)$$

where

$$V_B = \frac{1}{d_{np}} \sqrt{\frac{18 \kappa T}{\pi \rho_{np} d_{np}}}, \chi = \sqrt[3]{\frac{\pi}{6 \phi}} d_{np}, B = \mu_{bf}^{-1} [(c_1 d_{np} \times 10^9 + c_2) \phi + (c_3 d_{np} \times 10^9 + c_4)]$$

$$c_1 = 1.33 \times 10^{(-6)}; c_2 = 2.771 \times 10^{(-6)}; c_3 = 9 \times 10^{-8} \text{ and } c_4 = 3.93 \times 10^{-7}$$

The physical properties used in this modeling equation of the nanoparticles are listed in the Table 1. Over

TABLE 1 Physical properties of nanofluids.

Material	ρ (kg/m ³)	C_P (J/kg/K)
Water	998.96	4324
CuO	6350	535.6

TABLE 2 Domain independence test at $\text{Re} = 1$ for CuO-water nanofluids ($\phi = 0.03$, $d_{np} = 30$ nm).

D_∞/B	C_D	
	$\text{Da} = 1 \times 10^{-2}$	$\text{Da} = 1 \times 10^{-6}$
80	10.402	9.432
100	10.334	9.344
125	10.268	9.275



the range of temperature 300 K to 320 K, these properties assumed to be constant.

Finally, the solution of governing Equation (1)–(4) is completed with boundary conditions which have been expressed in dimensionless form,

At the inlet (uniform flow): $U = 1, V = 0$.

At the outlet: At the outlet, Neumann boundary condition and zero relative pressure are employed, i.e., $P = 0, \partial\psi/\partial X = 0$, where $\psi = U, V$.

At the porous-fluid interfaces^{14,23}: Pressure, velocity, temperature, shear stress and heat flux continuity is applied at the interface between fluid and porous regions. A constant wall temperature is employed on the cylinder's surface.

The specified dimensionless variables are obtained after the non-dimension of governing equations which is defined as $Re = \rho_{nf} V_{\infty} B / \mu_{nf}$ is the Re , $Da = K/B^2$ is the Darcy number, and $\Lambda = \mu_e / \mu_{nf}$ is the viscosity ratio. The viscosity ratio is considered to be equal to one in this study. It is important to introduce certain post-processing parameters that are utilized

in this study to analyze the numerical outcomes. For example,

Recirculation length (L_r): The non-dimensional length (L_r) from the rear stagnation point of porous square cylinder to reattachment point of the near streamline ($U = V = 0$) on the plane of symmetry ($X = 0$).

$$\text{Pressure coefficient}(C_p) : C_p = \frac{p_s - p_{\infty}}{1/2 \rho_{nf} V_{\infty}^2} \quad (5)$$

TABLE 3 Grid sensitivity test at $Re = 40$ for CuO-water nanofluids ($\phi = 0.03, d_{np} = 30$ nm).

Grid	δ	N_p	C_D	
			$Da = 1 \times 10^{-2}$	$Da = 1 \times 10^{-6}$
M3	0.0084	480	1.273	2.443
M2	0.0100	400	1.254	2.459
M1	0.0125	320	1.207	2.470

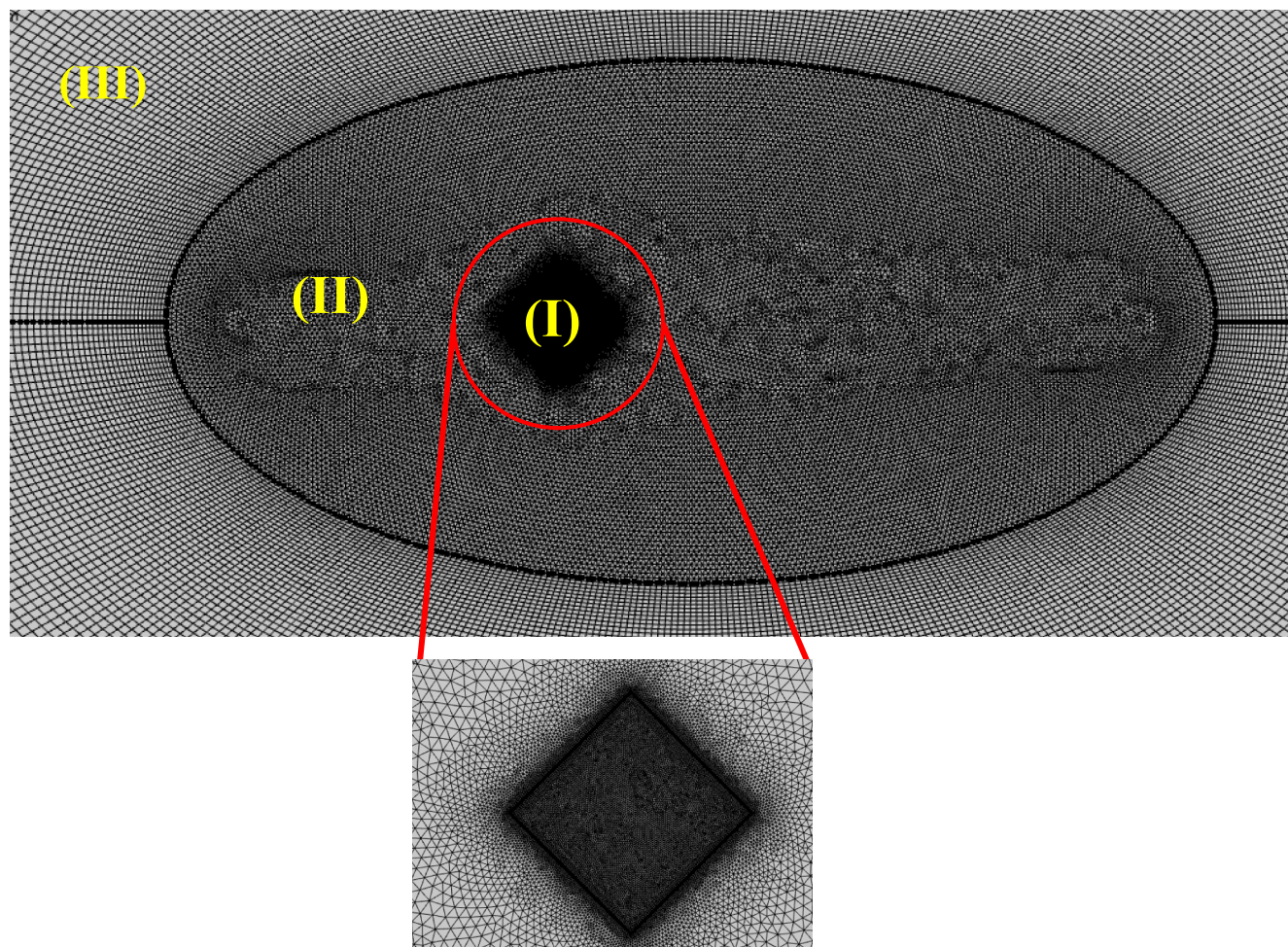


FIGURE 2 Typical grid structures near and through 45° inclined porous square cylinder.

$$\text{Drag Coefficient}(C_D) : C_D = \frac{F_D}{\frac{1}{2}\rho_{nf}V_\infty^2(\sqrt{2}B)} \quad (6)$$

The governing differentials Esq. (1)–(3), subject to the aforementioned boundary conditions, are eventually solved using the primitive variables $\Gamma(P, U, V)$. The numerical outcomes are then post-processed to assess the integral and local quantities, which are briefly described. The comprehensive numerical results presented in this study, characterized in the terms of pressure coefficient, vorticity and drag coefficient are anticipated to be dependent on Re , Da , and ϕ . The objective of this work is to investigate this functional relationship for the flow of nanofluids past a 45° inclined porous square cylinder.

3 | NUMERICAL DETAILS

The current problem of nanofluids flow over the 45° inclined porous square cylinder is converted into the governing differential Equations (1–3) along with applied boundary conditions. These equations are solved by using COMSOL (FEM based commercial software).^{24,25} The flow domain is discretized into the non-uniform

triangular and rectangular elements. The steady state, 2D, Laminar flow and Brinkman-Forchheimer equation along with (PARDISO) solver was used to solve the resulting system of non-linear equations. The relative convergence criterion of 10^{-5} was used for velocity and pressure field. Moreover, the user defined functions (UDF) are used to introduce the density and viscosity of nanofluids as well as the interface boundary condition between porous cylinder and nanofluids.

4 | DOMAIN AND GRID INDEPENDENT STUDY

The selection of the size of artificial computational domain is crucial parameters that directly impact the global parameters. The domain independence test is investigated at least value of Re . In this case, the hydrodynamics boundary layers are expected to be thick which thereby influencing on the size of the domain. Therefore, the domain independence test has been investigated at low value of Re and extreme value of Da and ϕ , as given in Table 2. Here, the circular imaginary domain of diameter D_∞ is developed to avoid some parameters such as upstream and downstream length. The numerical

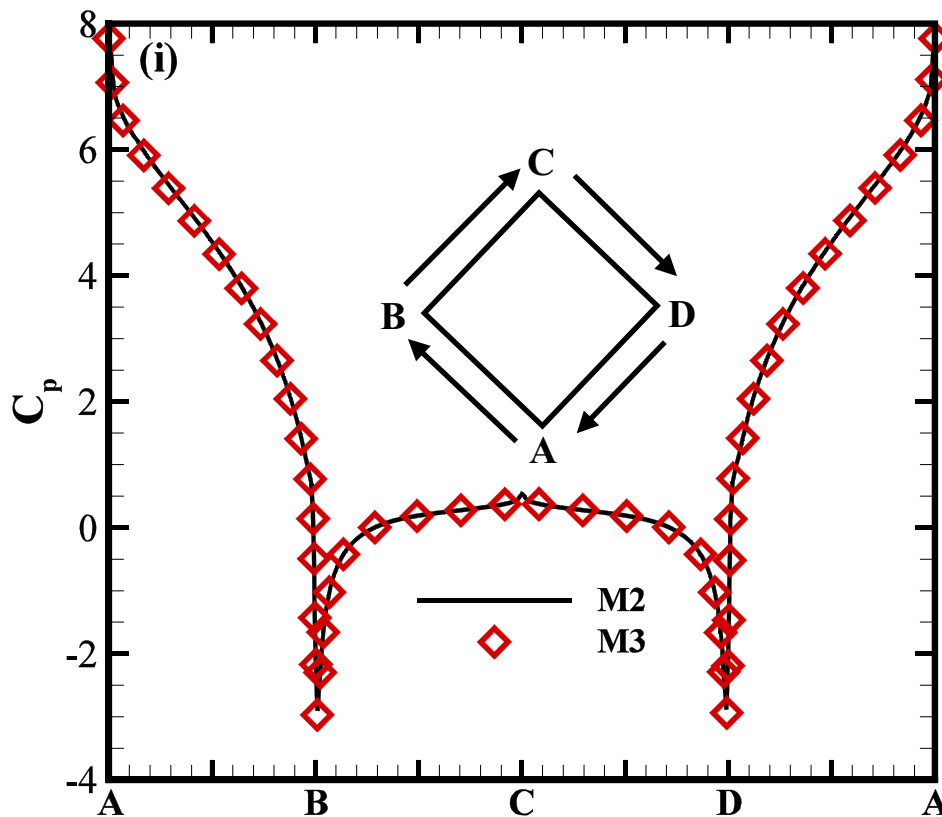


FIGURE 3 Grid sensitive analysis on pressures coefficient (C_p) over the surface of 45° inclined porous square porous cylinder at $Re = 40$, $\phi = 0$, $Da = 10^{-6}$.



simulation has been performed by changing the imaginary circular domain (D_∞) as $80B$, $100B$ and $125B$.^{24,26–28} A detail investigation of the results suggested that the variations in the value of global parameters such as C_D by changing the size of the domain from, $D_\infty = 100B$ to $D_\infty = 120B$ is negligible ($<1\%$). Therefore, the domain $D_\infty = 100B$ is selected as the optimum domain size for the present investigations.

After fixing the optimum domain size, the grid independence test is carried out to select the optimum grid size. The computational domain is divided into three sub-domain to minimize the computational time, i.e., sub-domain-I (45° inclined porous square cylinder), Sub-domain -II and Sub-domain - III, as shown in Figure 2. To capture steep pressure and velocity gradients, very fine grid size is required near as well as inside the 45° inclined porous square cylinder.

The three distinct grids namely, M1, M2 and M3 have selected by changing the number of elements on the porous cylinder surface, as presented in Table 3. The

grid independence study has been done for three different grids namely M1 ($\delta/B = 0.0084, N_p = 480$), M2 ($\delta/B = 0.01, N_p = 400$) and M3 ($\delta/B = 0.0.125, N_p = 320$) by changing the size of element and total number of elements on the porous cylinder.

In the result, the post processing parameters such as C_D is calculated for three different grid sizes. From the results, it is clearly observed that the negligible difference ($<1\%$) in the value of C_D when changing the grid from M2 to M3.

Thus, grid M2 is selected as optimum grid size. The selected grid M2 is further verified by plotting the pressure coefficient (C_p) variation on surface of the porous cylinder, as shown in Figure 3 for both grids (M2 and M3) and there is negligible difference between results. Therefore, the grid M2 is selected to be optimum grid for the current study. These numerical parameters are further manifested by presenting a validation results in the next section.

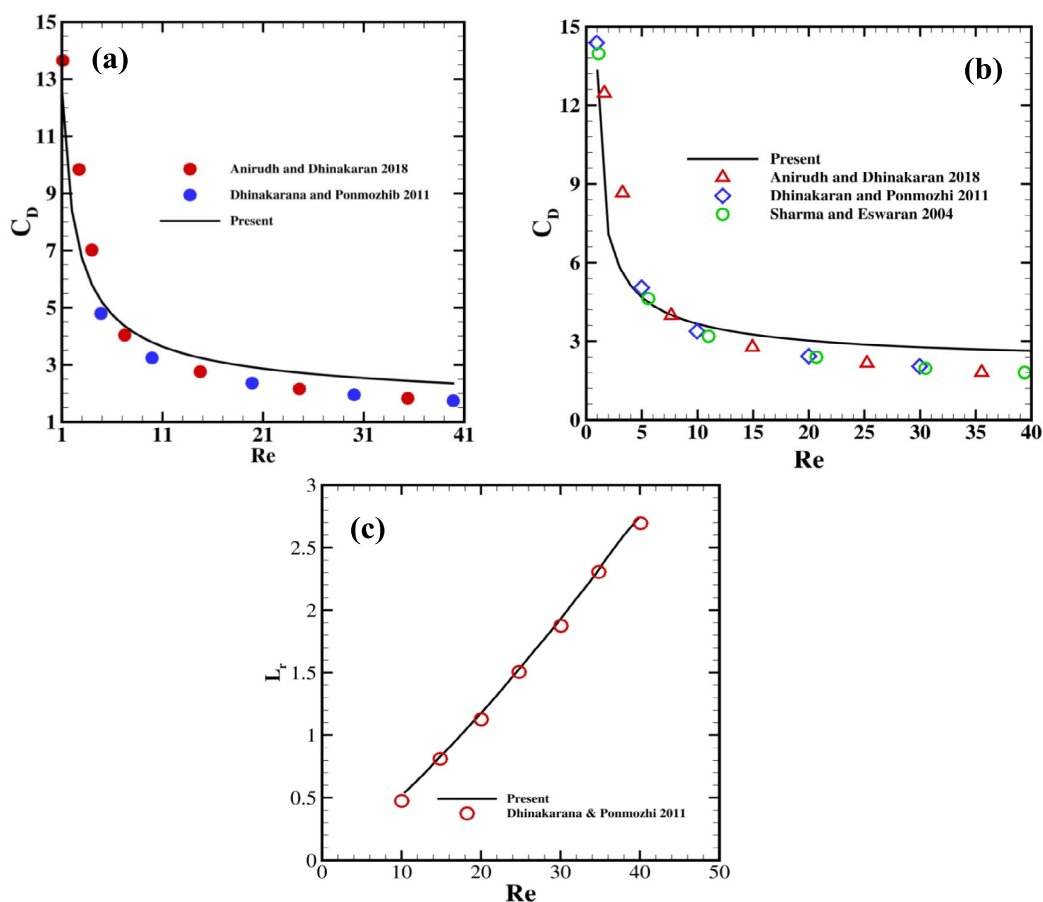


FIGURE 4 Validation results with square cylinder (a) C_D at $Da = 10^{-3}$ ($\phi = 0$), (b) C_D at $Da = 10^{-6}$ ($\phi = 0$) and (c) L_r at $Da = 10^{-3}$ ($\phi = 0$).



5 | VALIDATION OF PRESENT NUMERICAL METHOD

To ensure the accuracy of our numerical methodology, the code is validated by comparing it with previous studies of similar configurations. The flow over a porous square cylinder (0° inclined) is used as a validation purpose. Figure 4a,b shows the variation of drag coefficient with the Reynolds number at $Da = 10^{-3}$. Again an excellent agreement is observed with the results of Anirudha and Dhinakarana 2018^{29,30} and Dhinakarana & Ponmozhi 2011.⁷

In this case the maximum deviation is observed to be 2.8%. Furthermore, Figure 4c shows the recirculation length variation with Re at $Da = 10^{-3}$, and compared present numerical results with Dhinakarana & Ponmozhi 2011.⁷ An again excellent agreement is found between them and maximum deviation found to be less than 2%.

6 | RESULTS AND DISCUSSION

The aim of the current study is investigate the effect of Re , Da , ϕ on various hydrodynamics characteristics such as velocity fields, recirculation length, vorticity and drag coefficient. The current study is restricted to 2D, steady and laminar flow up to $Re = 40$ and spherical

nanoparticle ($d_{np} = 30\text{nm}$) in CuO-water nanofluids. In this case, numerical results are performed in the ranges of parameters, $1 \leq Re \leq 40$, $10^{-2} \leq Da \leq 10^{-6}$, $0 \leq \phi \leq 0.05$ for single porosity ($\varepsilon = 0.9$).

6.1 | Streamline profiles and velocity distribution near and inside the porous square cylinder

Figure 5a–c illustrates the streamline profiles near a 45° inclined porous square cylinder within an unconfined domain. These profiles serve to identify the stagnant and recirculation zones at various Darcy number values ($Da = 10^{-2}$, $Da = 10^{-4}$ and $Da = 10^{-6}$). The impact of Reynolds number (Re) on two distinct nanoparticle concentrations or volume fractions, $\phi = 0$ (base fluid) and $\phi = 0.03$ (nanofluids), is depicted in the results.

Additionally, in the case of a porous cylinder, the Darcy number (cylinder permeability) is crucial in determining the velocity profile of the fluid surrounding the 45° inclined porous square cylinder. The figure displays streamline profiles at various values of Re , Da , and ϕ showcasing the interactions between fluid flows and the porous media (square cylinder), resulting in diverse flow behaviors. In Figure 5a, when $Da = 10^{-2}$, no flow

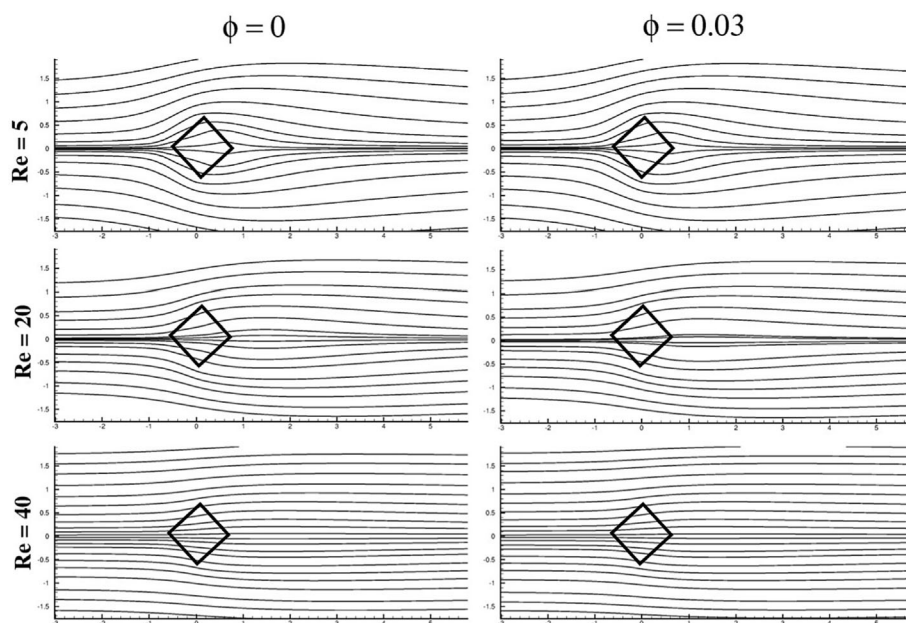


FIGURE 5 (a) Variation of streamline profiles through and around a 45° inclined porous square cylinder for different value of Re , ϕ at $Da = 10^{-2}$. (b) Variation of streamline profiles through and around a 45° inclined porous square cylinder for different value of Re and ϕ at $Da = 1 \times 10^{-4}$. (c) Variation of streamline profiles around a 45° inclined porous square cylinder for different value of Re and ϕ at $Da = 10^{-6}$.

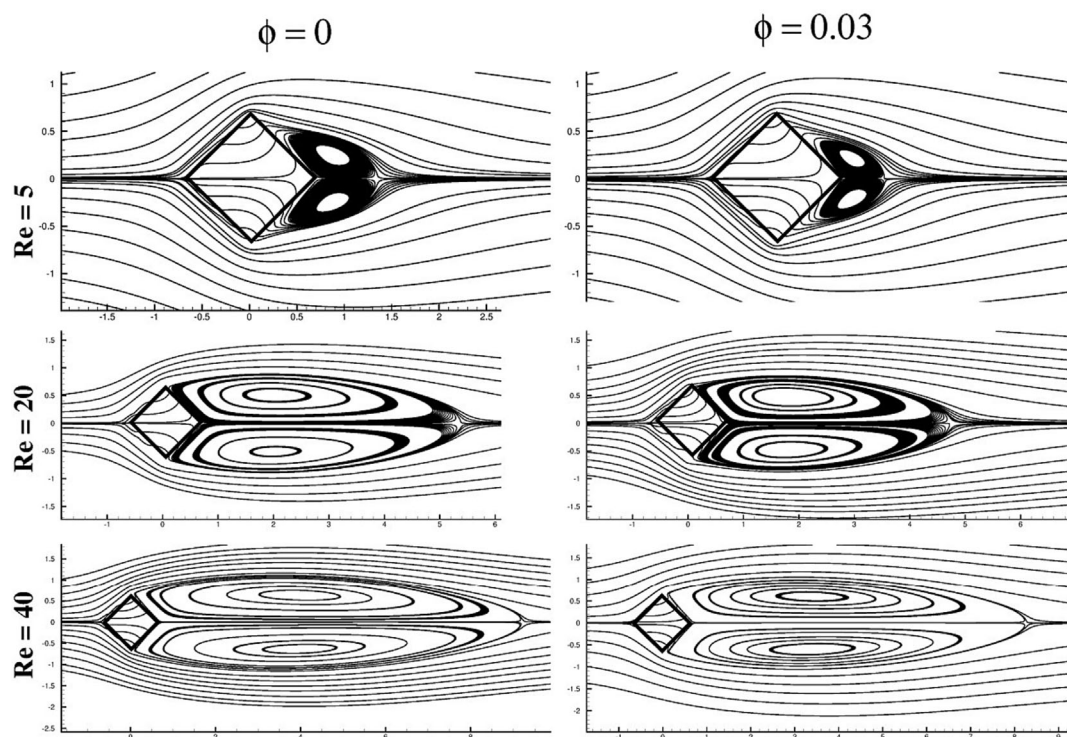


FIGURE 5 (Continued)

separation are observed at the rear of the square cylinder across all values of Re and ϕ . Additionally, a significant volume of fluid passes through the porous square cylinder due to its high permeability. At this Darcy number, the porous cylinder offers minimal resistance to the penetrating fluid, resulting in only minor deviations in the downstream streamlines. A discernible variation in streamlines becomes apparent when Reynolds numbers are low, indicating a reduced prevalence of inertial forces. Conversely, the impact of nanoparticle volume fraction on streamline profiles has a negligible effect at $Da = 10^{-2}$, as clearly observed.

On further decreasing the permeability level, such as $Da = 10^{-4}$ in Figure 5b, there is a slight reduction in the quantity of fluid penetrating the porous cylinder, resulting in a reduction in the resistance force offered by the cylinder.

Consequently, adverse pressure gradient leads to the formation of recirculation zones at the rear side of the square cylinder.^{7,29}

It is observed that flow separation increases with Re , while it decreases with the ϕ for nanofluids with a diameter size of 30 nm. The decrease in flow separation with ϕ is attributed to the lower effective Reynolds number based on nanofluids compared to clear water.^{25,26,31}

As an example, at $\phi = 0.03$, the viscosity of nanofluids increased by 23%, while the density increased by

53% while maintaining constant values of Re and Da . The incorporation of nanoparticles to the base fluids (or increase in ϕ) resulted in the dominant influence of viscous forces in nanofluids, thereby decreasing the effective Re . Next, Figure 5c depicted the streamline profile at extreme value of $Da = 10^{-6}$.

For lower value of Re , the flow is entirely attached to the cylinder and follow the shape of the cylinder due to the lower inertia force on the cylinder's surface.

In this case, the inertial forces become dominant over the viscous forces, and hence, recirculation zones formed at the rear side of the cylinder, which is consistent with Kaur et al.²⁵ for 45° inclined solid square cylinder. This is due to the fact the flow resistance is high at a lower value of Darcy number, which allows less or no fluid to flow through it, which results in increasing recirculation zones near the cylinder, which is evident. There is slight reduction in streamline profiles with the addition of nanoparticles. It is visualized the effect of nanoparticles volume fraction in Figure 5c.

Furthermore, an interesting behavior is observed at intermediate value of $Da = 10^{-3}$ to $Da = 10^{-2}$ as shown in Figure 6. At particular value of $Re = 10$ and $\phi = 0.01$ for Darcy number, $Da = 10^{-3}$, two counter rotating separation bubbles are formed at downstream side of square cylinder, some portion of the separation bubbles are attached to the triangular after body of 45° inclined porous square cylinder. On increasing the Darcy number

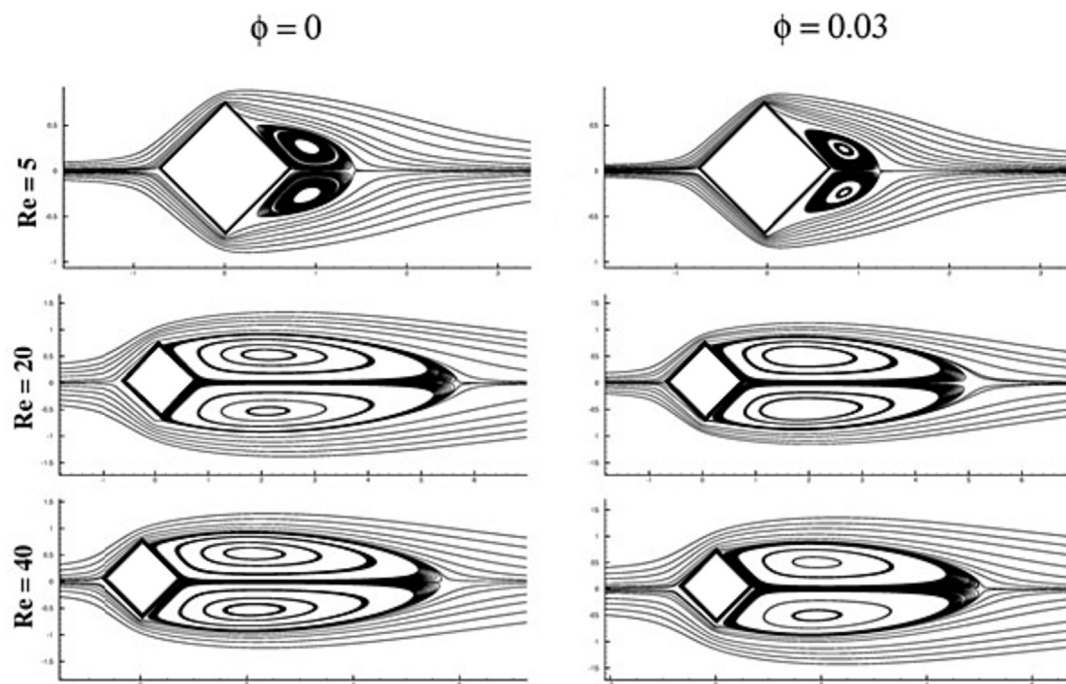


FIGURE 5 (Continued)

such as $Da = 7 \times 10^{-3}$, flow detached from the triangular after body of 45° inclined porous square cylinder, the separation bubbles become smaller due to the increasing permeability of porous cylinder. On further increasing the Darcy number (such as $Da = 6 \times 10^{-3}$), the separation bubbles completely disappeared due maximum permeability of cylinder and strong inertial dominated effect which is denoted as critical Darcy number (Da_c).

It is found that the value of Da_c is depended on the value of Re and ϕ . The values of Da_c are calculated at different value of Re and ϕ as given in Table 4.

It is clearly indicated that the value of Da_c changes positively with ϕ and inverse dependent with Re . At low value of volume fraction, $\phi = 0.01$, the separation zone detached from the 45° inclined porous square cylinder and disappeared at $Da_c = 5.5 \times 10^{-3}$ whereas for $\phi = 0.05$, the separation zone attached and disappeared at $Da_c = 6 \times 10^{-3}$. These kind of interesting behaviors are only observed at intermediate value of Darcy number ($Da = 10^{-3}$ to $Da = 10^{-2}$) due to the interplay between inertia force of fluid and permeability of cylinder.

Figure 7 illustrates the alteration of U velocity along the horizontal direction for different value of Da at $Re = 30$. In this case, the origin of the 45° inclined porous square cylinder is located at the center of the porous cylinder. It is clearly seen that at small value of Darcy number ($Da = 10^{-6}$), the U -velocity approaches to zero inside the porous square cylinder due to no penetration

boundary condition at the surface of the cylinder and large resistant against the penetrating fluid flow. At Darcy number ($Da > 10^{-3}$), the velocity (U) distribution inside the porous cylinder is non-zero. The velocity variation increases up to 32% of free stream velocity due to the high permeable cylinder and large amount of fluids passing through it. Figure 7 shows the velocity distribution in the x -direction ($y = 0$) for different volume fraction ($\phi = 0.01, 0.03$ and 0.05), at $Re = 30$ and $Da = 10^{-2}$.

From this figure, it is obtained that the velocity distribution inside the porous cylinder decreases with addition of nanoparticle in base fluids due to the lower effective Reynolds number of nanofluids.

6.2 | Variation of recirculation length

In general, the recirculation length measured the size of the flow separation region that forms in a flow when a fluid encountered a 45° inclined porous square cylinder. Figure 8a depicts the variation of the recirculation length (L_r) with Re for different ϕ at $Da = 5 \times 10^{-3}$. It is clearly seen that at constant value of Da and ϕ , the value of L_r gradually enhances as the value of Re increases due to gradual dominance of the inertia strength over the viscous strength. At high Reynolds numbers (such as $Re = 40$), the flow separation region is found to be larger, and the recirculation length increased significantly. However, it is also noted that the complex relationship

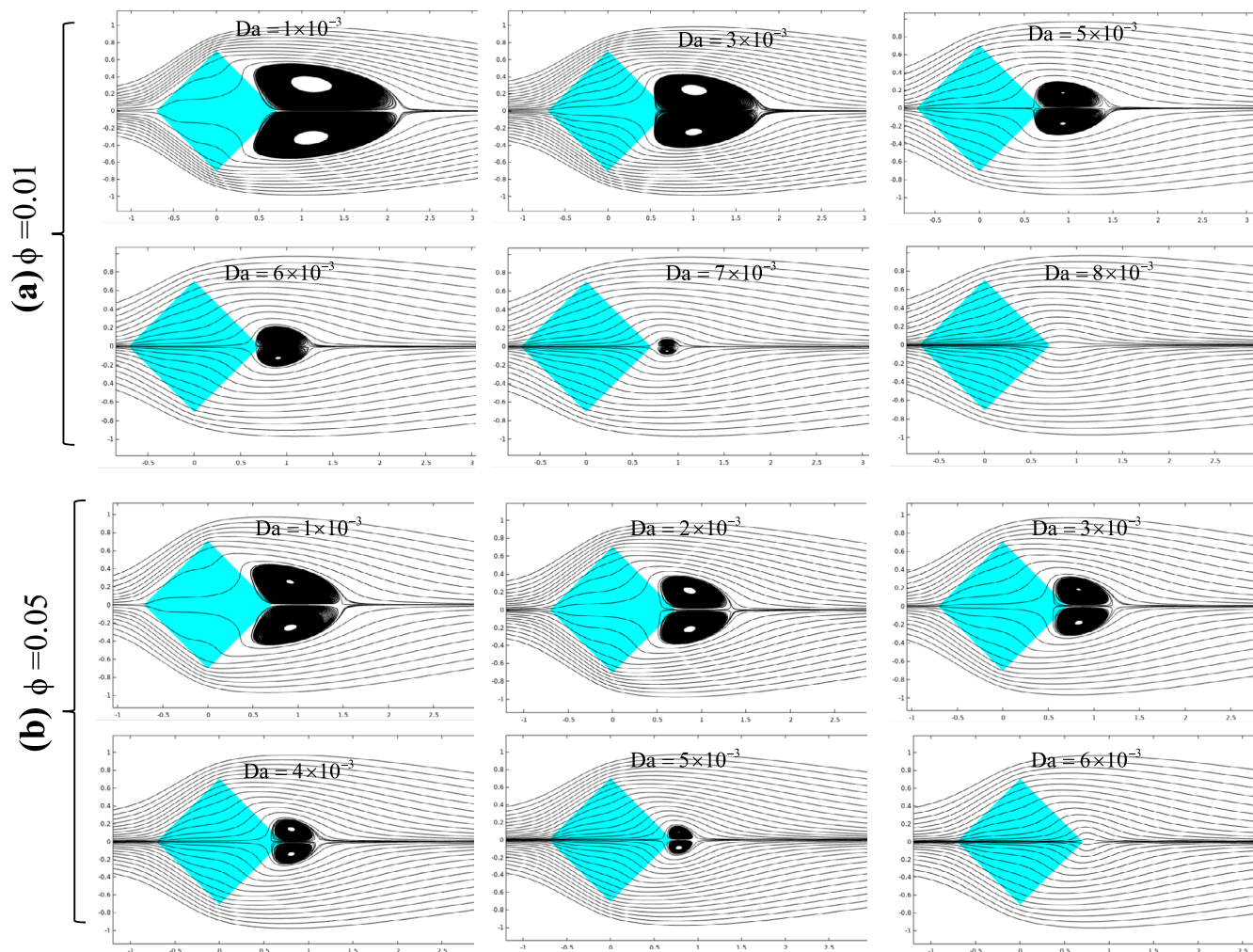


FIGURE 6 Flow fields near and through the porous cylinder at intermediate value of Darcy number at (a) $\phi = 0.01$. (b) $\phi = 0.05$ at $Re = 10$.

TABLE 4 Critical Darcy number (Da_c) at different value of Reynolds number (Re).

Re	$\phi = 0.01$	$\phi = 0.03$	$\phi = 0.05$
10	7.4×10^{-3}	7.3×10^{-3}	5.9×10^{-3}
20	7.0×10^{-3}	7.1×10^{-3}	7.7×10^{-3}
30	6.3×10^{-3}	6.4×10^{-3}	7.8×10^{-3}
40	5.5×10^{-3}	5.7×10^{-3}	6.0×10^{-3}

between L_r and Re changing nonlinearly with Re (i.e., expected trend of $L_r = a_1 Re^2 + b_1 Re$, where a_1 and b_1 are the constant value). In Figure 8b, it is indicated the variation of L_r with ϕ at different value of Da irrespective of Re .

As expected, the increasing nanoparticle volume fraction in nanofluids resulted in a reduction of the recirculation length. With increasing the value of ϕ subjected the more viscous strength with respect to inertial strength,

which is altered the effective Re , change the flow separation region. For instance, the increases value of L_r is found to be 54.67% when increases the nanoparticle concentration from $\phi = 0$ (base fluid) to $\phi = 0.05$ (nanofluids) at $Re = 30$ and $Da = 5 \times 10^{-3}$. The expected trend of L_r with nanoparticle volume fraction is indicated as $L_r = -a_2 \phi^2 + b_2 \phi + c_2$, where a_2 , b_2 and c_2 are the constant values. The variation of L_r with ϕ is affected by the value of Da (except $Da = 1 \times 10^{-2}$) and Re .

Figure 8c shows the variation of L_r with Da at $Re = 30$ for two different value of $\phi = 0$ (base fluid) and $\phi = 0.03$ (nanofluids). It is found that the recirculation length decreases with increasing value of Da . This is because of higher value of Darcy number indicates the porous medium is more permeable, which means that the fluid can flow through it more easily. As a result, the flow rear side of the porous square cylinder is less likely to separate, leading to a shorter recirculation length. Conversely, as the value of Da decreases, the

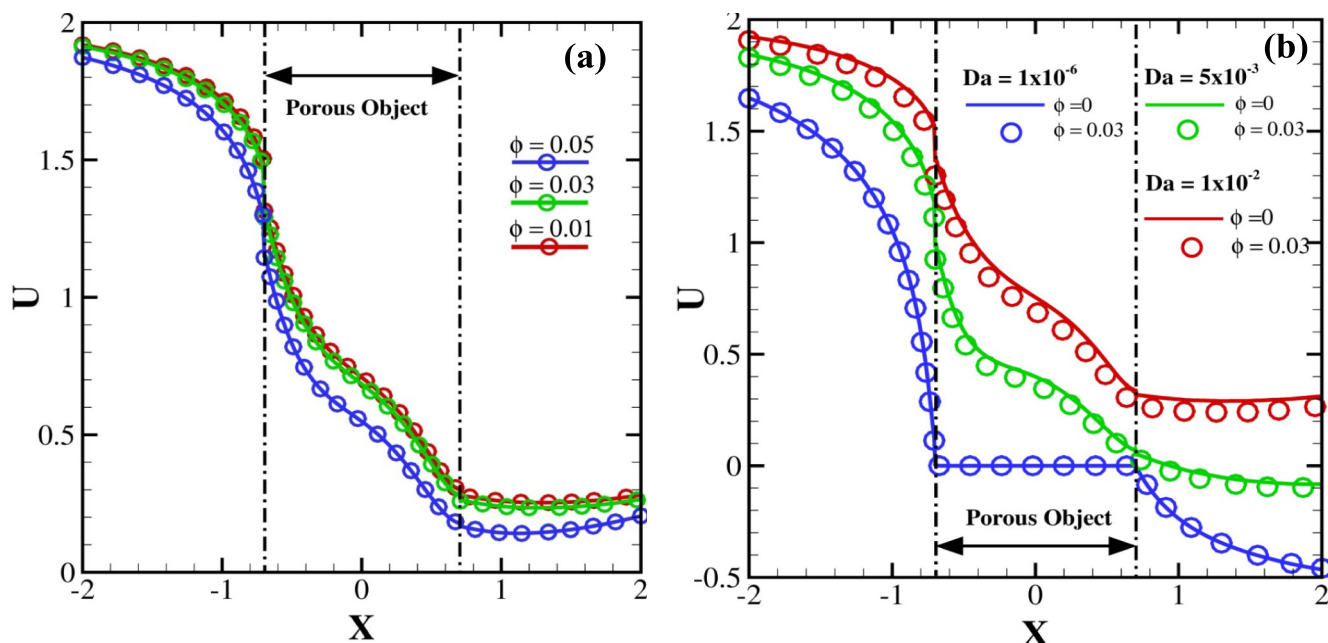


FIGURE 7 Velocity distribution (U) in the x -direction ($y = 0$) for different Darcy number at $Re = 30$.

recirculation length typically increases. It can be seen that recirculation length for porous square cylinder approaches asymptote, which is similar to the 45° inclined solid square cylinder. The predicted trend of L_r with Da is expected the exponential curve ($L_r = a_3 e^{-b_3 Da}$, where a_2 and b_2 are the constant values), as shown in Figure 8c. The value of L_r can be varied by changing the value of Re , Da and ϕ .

6.3 | Vorticity contours

Vorticity contours refer to the flow pattern of fluid around a cylinder. The contours are a measure of the strength and direction of the fluid's rotational motion. The existence of a square cylinder causes the fluid to flow in a complex manner, leading to the creation of vortices and eddies. Figure 9 represents the vorticity contours near the porous square cylinder by varying the value of Re , Da and ϕ at $Da = 10^{-6}$ and $\phi = 0.01$, the vorticity contour are not able to flow inside the square cylinder due to the solid and non permeable behaviors of cylinder at every value of Re . It is clearly observed that higher magnitude strength appears in upper and lower corner of the cylinder. It is also evident the vorticity zone near the upper and lower corners rises and shifts towards the rear side of the cylinder as Re increases. Simultaneously, the size of vorticity zone in the rear side increases with Re and leads to the rise the recirculation zone. On the other hand, as the concentration of nanoparticle increases from $\phi = 0.01$ to $\phi = 0.05$, the vorticity

zone slightly decreases in their size due to increasing the recirculation length at the downstream side of the 45° inclined porous square cylinder.

Additionally, as the Darcy number (Da) increases from a value of $Da = 10^{-6}$ to $Da = 10^{-3}$, it is observed that the vorticity contours penetrate into the cylinder for all Re . The vorticity zone located at the upper corner of the cylinder undergoes stretching and a shift towards the rear side of the cylinder as Re increases. The permeable nature of the porous cylinder results in the vorticity zones penetrating the afterbody and forebody shapes of the 45° degree inclined square cylinder.

The impact of altering the volume fraction of nano-fluid within the system is also found to be significant. Further increases in the value of Da from $Da = 10^{-3}$ to $Da = 10^{-2}$, at a volume fraction of 0.01, result in an enhancement of the strength of the vorticity contours and a shift towards the flow direction. The penetration of vorticity inside the porous cylinder is greater compared to previous cases ($Da = 10^{-3}$ and $Da = 10^{-6}$). As a result, the reduction in the size of the vorticity zones has been observed and the recirculation zone at the downstream side of the cylinder disappears in the corresponding streamline profiles. When the ϕ is increased from 0.01 to 0.05, a reduction in the strength of the vorticity zones is visualized. The effect of the inertial force by the nanofluid and the permeability of the 45° inclined porous square cylinder play a significant role in the flow fields. It is clearly seen the effect of various value of Re , Da and ϕ in the vorticity contours in the flow fields.

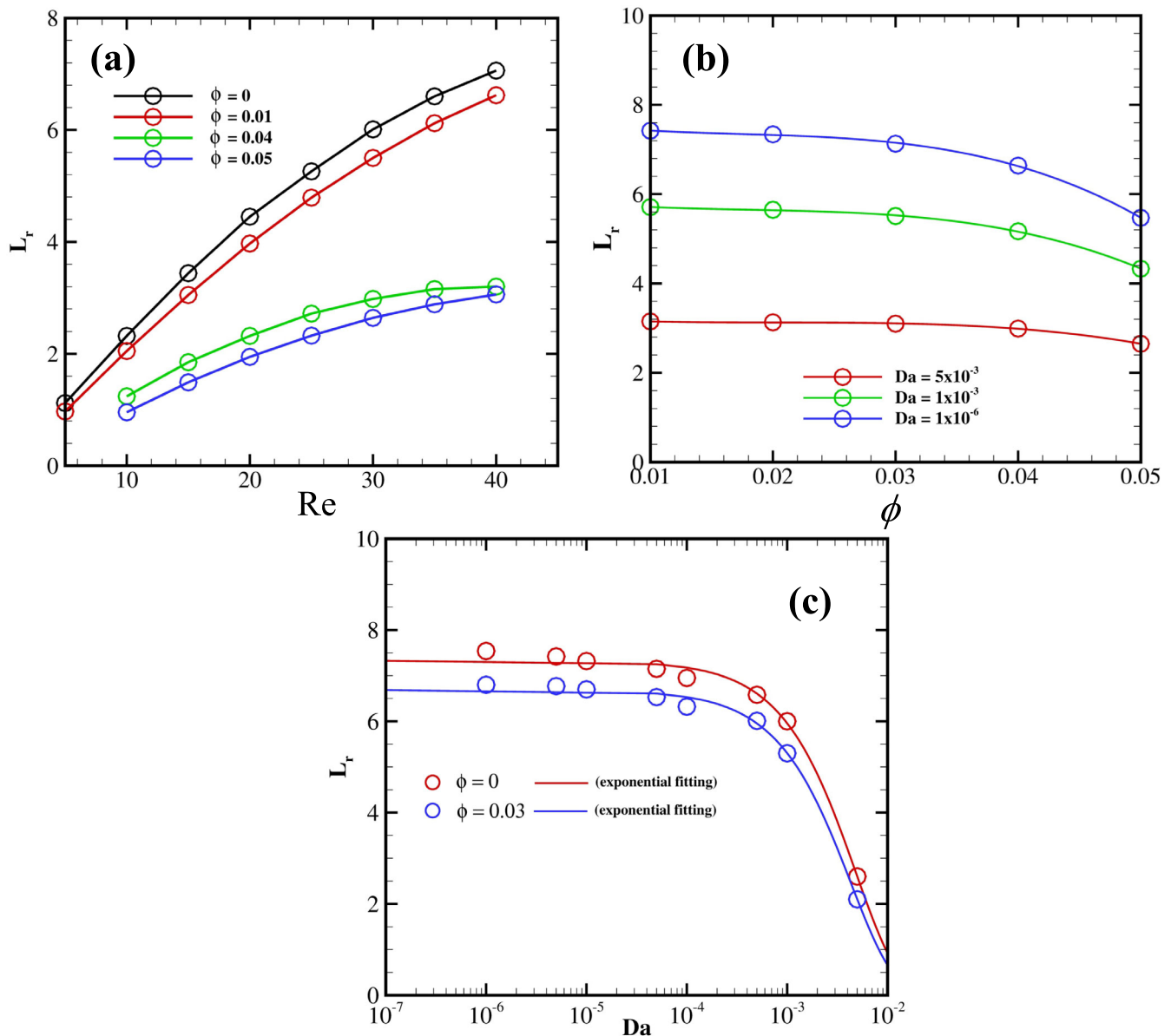


FIGURE 8 Variation of recirculation length with Re , ϕ and Da .

6.4 | Pressure coefficient

Figure 10 shows the local change of pressure coefficient on the 45° inclined square cylinder surface by changing the value of ϕ , Re and Da . In Figure 10, the variation of C_p plotted for the case $\phi = 0.01$, at $Re = 1$ and $Da = 10^{-6}$, the maximum value of C_p occurred on front stagnation point-a, and then the value of C_p reaches to the corner point-b after that suddenly drop to the negative value of C_p , and then again increases to the maximum value of C_p up to the rear point-c. The variation trend of C_p at $Da = 10^{-6}$ is consistent with the variation of C_p on the 45° inclined solid square cylinder.^{25,32} As the value of Da increases such as $Da = 10^{-4}$, $Da = 10^{-3}$ and $Da = 10^{-2}$, ultimately increasing the permeability of

cylinder and more fluids passing through the cylinder, the variation of C_p is found to be decreasing, especially near the corner point-b. As the Reynolds number increases, $Re = 10$, significantly increasing the value of inertial strength, which thereby decreasing the strength of the pressure coefficient. On further increasing the Reynolds number, $Re = 40$, similar trend of C_p is observed except, $Da = 10^{-2}$ (almost all fluids pass through the cylinder) and the strength of the pressure coefficient is reduced due to the higher inertial force. Figure 10 indicated the variation C_p on the square cylinder for $\phi = 0.05$, it is expected that the variation of C_p is higher as compare to the $\phi = 0.01$.^{25,26,31} The variation trend of C_p would remain similar with the value of Da and Re . From the figure, it is clear that the value of C_p

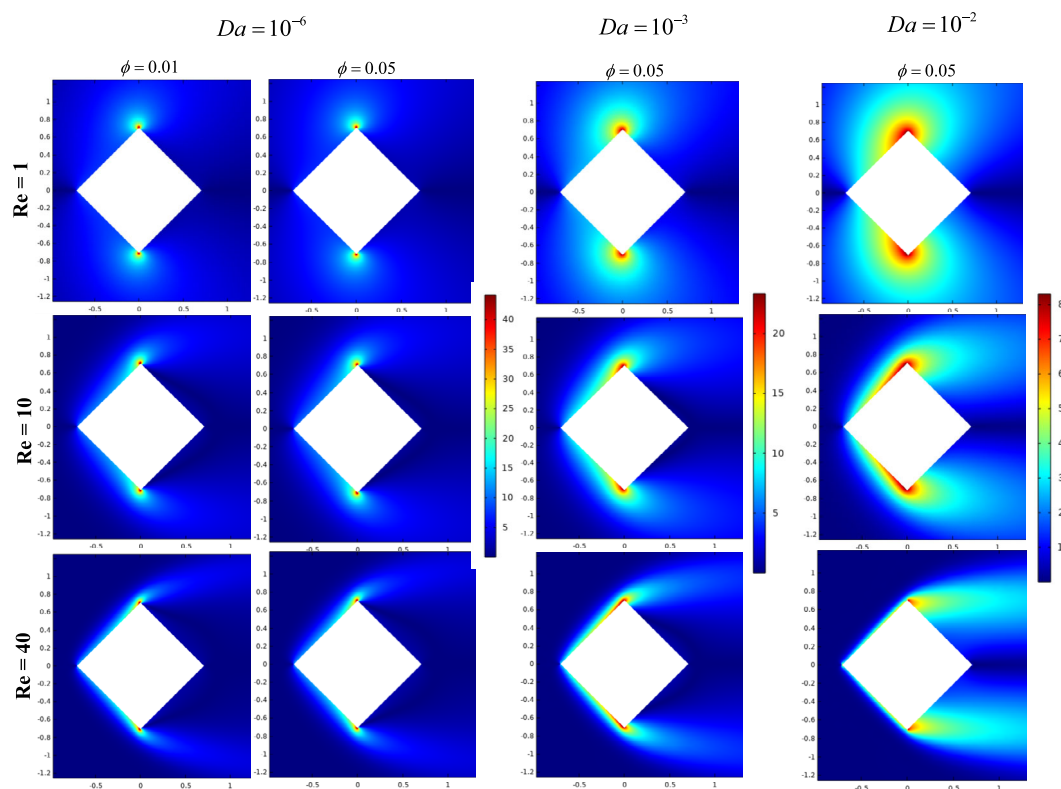


FIGURE 9 Variation of vorticity contours at Re , Da and ϕ .

has shown positive variation with nanoparticle concentration for all value of Re and Da .

6.5 | Drag coefficient

The net hydrodynamic force applied by nanofluid on 45° inclined porous square cylinder is represented in terms of the total drag coefficient (C_D) which is defined as Equation 6. The value of C_D for clear fluids ($\phi = 0$) and nanofluid ($\phi = 0.05$) are plotted and compared the value of C_D at different value of Re and Da . From Figure 11, it is found that the variation of the C_D with Da is higher for the case of clear fluids ($\phi = 0$) as compared with the nanofluid ($\phi = 0.05$). Obviously, with addition of nanoparticle in clear fluid, the effective Reynolds number (based on ρ_{nf} and μ_{nf}) would be higher. As shown in the figure, the variation of C_D with Re indicates the classical inverses relationship (i.e., $Re \propto 1/C_D$) and consistent with the previous literature results.^{25,26,31} Irrespective of the value of Re and nanoparticle volume fraction (ϕ), the variations of C_D with Da is divided into two parts (i) at $10^{-6} \leq Da \leq 10^{-4}$, not significant change is observed in C_D due to the very small permeability of the 45° inclined porous square cylinder, and the value of C_D is almost same as the C_D of solid square cylinder, (ii) at $10^{-4} \leq Da \leq 10^{-2}$, the value of C_D is decreases with

increasing value of Da , more amount of fluids flow through the cylinder, which is also indicate the increasing permeability of 45° inclined porous square cylinder.

The value of C_D increases with the addition of nanoparticles in the water (with an increasing value of ϕ) at a fixed value of Da and Re . This is happened due to the viscosity and density of the nanofluids increasing with the addition of nanoparticles in base fluids.²⁵ In this case, the value of inertia forces is significantly reduced with respect to viscous forces thereby leading to the value of C_D with nanoparticle volume fraction (ϕ). Additionally, when a nanofluid with a high concentration of nanoparticle ($\phi = 0.05$) flows through a 45° inclined porous square cylinder having pores inside, it offers some kind of resistance to flow which results in a high value of drag coefficient as compared to base fluid ($\phi = 0$).

Finally, the comparison of the flow field over and through the square cylinder for two different orientation angle (0° and 45°) at $Re = 20$, $Da = 10^{-3}$ and $\phi = 0.05$, as shown in Figure 12. The flow separation length or recirculation length in the case of a square cylinder (0° inclined) is lower compared to that of a 45° inclined porous square cylinder. Despite having similar sharp edge features, the wake formation at downstream side of the cylinder is different due to upstream and downstream face of the cylinder. In the 0° inclined porous square

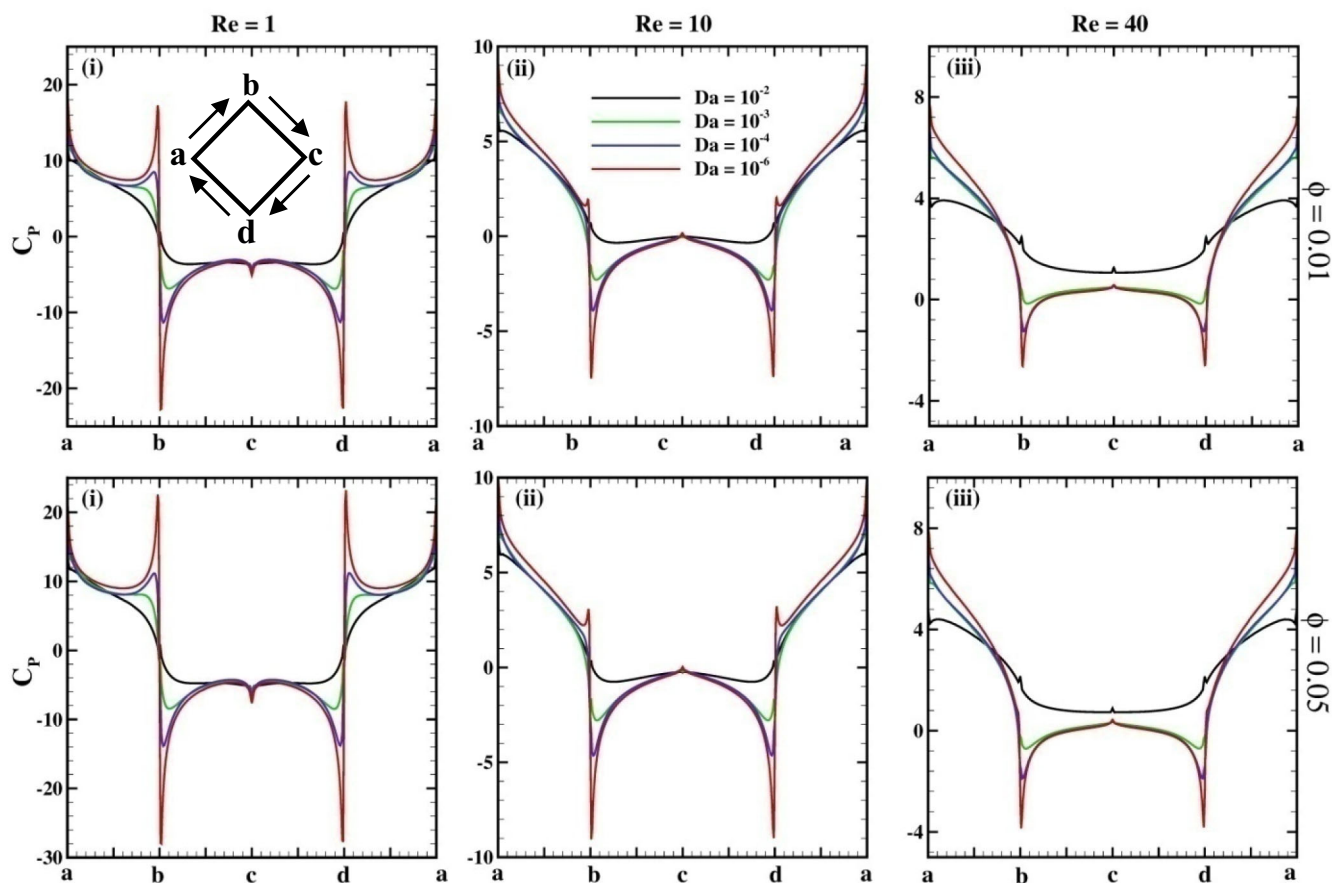


FIGURE 10 Variation of pressure coefficient on the surface of the porous 45° inclined square cylinder at different value Re , Da and ϕ .

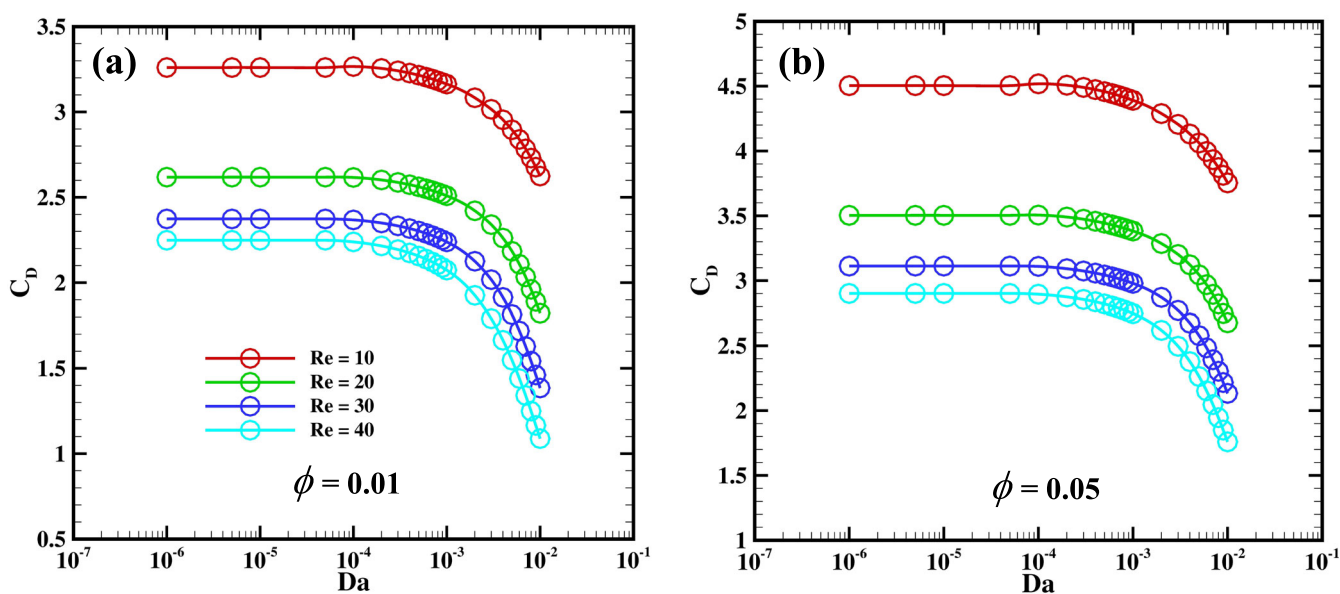


FIGURE 11 Variation of drag coefficient with Reynolds for 45° inclined porous square cylinder (a) at $\phi = 0.01$ and (b) at $\phi = 0.05$.

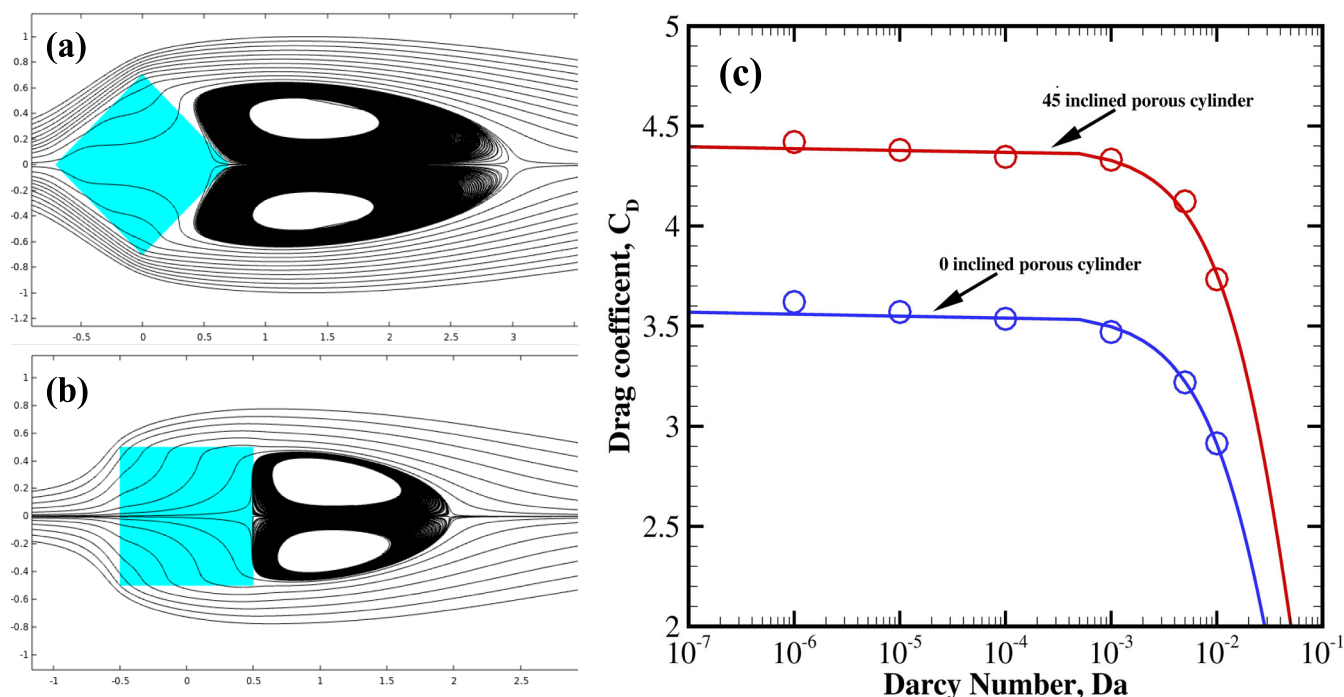


FIGURE 12 (a) Streamline profiles near and through 45° inclined square cylinder, (b) streamline profiles near and through 0° inclined square cylinder, (c) variation of drag coefficient with Darcy number for nanofluids.

cylinder having flat face at upstream and downstream side of the cylinder whereas the 45° inclined porous square cylinder has triangular face at upstream and downstream side. Therefore, the length and shape of flow separation are expected to be different which thereby affected the flow field. The flow separation attached to the rear side of the face at smaller value of Da however, with larger value of Da , more fluid passes through the porous cylinder, the separation zone detached and moves away from the rear side of the cylinder and also vanishes eventually. The different flow behaviors predicted the different drag coefficient which is directly related with flow fields. Figure 12c showed the variation of drag coefficient with Darcy number for both orientations. Here, we can clearly observe the drag coefficient for 45° inclined porous square cylinder is enhanced as compare to 0° inclined porous square cylinders. Therefore, the orientation angle is also an important parameter for controlling the hydrodynamics results, which we will include in future work.

7 | CONCLUSION

A numerical analysis of nanofluid flow over 45° inclined porous square cylinders has been conducted to explore the effect of ϕ , Da , and Re on the steady flow range. The effect of these parameters on streamline, velocity profiles inside the porous cylinder, recirculation length, pressure

coefficient, and drag coefficient were investigated. The numerical results are summarized as follows,

- A steady separation zone is attached to the rear side at a smaller value of Da ($=10^{-6}$); however, with a larger value of Da ($=10^{-2}$), more fluid passes through the porous cylinder, causing the separation zone to detach and move away from the rear side of the cylinder, eventually vanishing. In the existence of nanofluids, the size of the separation zone starts decreasing with an increase in nanoparticle volume fraction.
- The recirculation zone showed a positive correlation with Re and an inverse dependence on ϕ and Da . The critical Darcy number (Da_c) decreases with an increase in nanoparticle volume fraction at a fixed value of Re , while Da_c increases with Re .
- The variation in the pressure coefficient (C_p) on the surface of the porous cylinder indicates a positive dependency with Re , whereas C_p decreases with increment in the values of Da and ϕ .
- The value of the drag coefficient is almost constant up to a Darcy number of 10^{-5} , and a decreasing trend is observed between Darcy numbers of 10^{-5} and 10^{-2} . The drag coefficient is enhanced by reducing the Re and ϕ . Finally, we compared the flow field and drag coefficient of the present orientation of the porous square cylinder with a 0° inclined square cylinder, and found that the drag coefficient is higher for the present porous square cylinder (at a 45° inclination).



NOMENCLATURE

B	side of the square cylinder, m
C_D	drag coefficient (dimensionless), $C_D = \frac{F_D}{\frac{1}{2}\rho_{nf}V_\infty^2(\sqrt{2}B)}$
C_P	pressure coefficient (dimensionless), $C_P = \frac{P_s - P_\infty}{\frac{1}{2}\rho_{nf}V_\infty^2}$
Da	Darcy number (dimensionless), $Da = K/B^2$
d_{np}	diameter of nanoparticle, nm
F_D	total hydrodynamics forces, N
K	permeability of the cylinder
L_r	recirculation length, m
N_P	total elements at the porous cylinder
p	pressure, N/m ²
Re	Reynolds number (dimensionless), $Re = \rho_{nf}V_\infty B/\mu_{nf}$
T	temperature, K
U	velocity components (dimensionless), $U = \frac{V_x}{V_\infty}, V = \frac{V_y}{V_\infty}$
X, Y	x, y -components (dimensionless), $X = \frac{x}{B}, Y = \frac{y}{B}$

GREEK WORDS

ρ	density of fluid
ϕ	nanoparticle volume fraction
μ	viscosity of fluids
ε	porosity
Λ	viscosity ratio, $\Lambda = \mu_e/\mu_{nf}$
δ	minimum grid size on the surface of the cylinder, $\delta = \frac{B}{N_p}$

SUBSCRIPTS

e	effective
nf	nanofluids
np	nanoparticles
s	surface
∞	free stream

AUTHOR CONTRIBUTIONS

Jaspinder Kaur: Conceptualization; methodology; writing—original draft preparation. **Jatinder Kumar Ratan:** Supervision; writing—reviewing and editing. **Roderick Melnik:** Reviewing and editing. **Anurag Kumar Tiwari:** Supervision; writing—reviewing and editing.

ACKNOWLEDGEMENTS

We would like to thank IIT Ropar for the supports of computational facility.

CONFLICT OF INTEREST STATEMENT

The authors have no conflicts of interest to declare. All co-authors have seen and agree with the contents of the manuscript and there is no financial interest to report. I certify that the submission is original work and is not under review at any other publication.

ORCID

Jaspinder Kaur <https://orcid.org/0000-0002-6347-7936>
 Jatinder Kumar Ratan <https://orcid.org/0000-0002-0804-816X>
 Roderick Melnik <https://orcid.org/0000-0002-1560-6684>
 Anurag Kumar Tiwari <https://orcid.org/0000-0002-1360-1388>

REFERENCES

- Nield DA, Bejan A. *Convection in porous media*. Springer; 2006.
- Bear J. *Dynamics of fluids in porous media*. Dover; 1988.
- Bastian P, Kraus J, Scheichl R, Wheeler M. *Simulation of flow in porous media: applications in energy and environment*. De Gruyter; 2013. doi:10.1515/9783110282245
- Bhattacharyya S, Dhinakaran S, Khalili A. Fluid motion around and through a porous cylinder. *Chem Eng Sci*. 2006; 61(13):4451-4461. doi:10.1016/j.ces.2006.02.012
- Yu P, Zeng Y, Lee TS, Chen XB, Low HT. Steady flow around and through a permeable circular cylinder. *Comput Fluids*. 2011;42(1):1-12. doi:10.1016/j.compfluid.2010.09.040
- Chen X, Yu P, Winoto SH, Low HT. Numerical analysis for the flow past a porous square cylinder based on the stress jump interfacial conditions. *Int J Numer Methods Heat Fluid Flow*. 2008;18(5):635-655. doi:10.1108/09615530810879756
- Dhinakaran S, Ponmozhi J. Heat transfer from a permeable square cylinder to a flowing fluid. *Energ Conver Manage*. 2011; 52(5):2170-2182. doi:10.1016/j.enconman.2010.12.027
- Valipour MS, Rashidi S, Bovand M, Masoodi R. Numerical modeling of flow around and through a porous cylinder with diamond cross section. *Eur J Mech - B/Fluids*. 2014;46:74-81. doi:10.1016/j.euromechflu.2013.12.007
- Vijayababu TR, Anirudh K, Dhinakaran S. Mixed convective heat transfer from a permeable square cylinder: a lattice Boltzmann analysis. *Int J Heat Mass Transf*. 2017;115:854-870. doi:10.1016/j.ijheatmasstransfer.2017.08.033
- Etminan-Farooji V, Ebrahimnia-Bajestan E, Niazmand H, Wongwises S. Unconfined laminar nanofluid flow and heat transfer around a square cylinder. *Int J Heat Mass Transf*. 2012; 55(5-6):1475-1485. doi:10.1016/j.ijheatmasstransfer.2011.10.030
- Valipour MS, Ghadi AZ. Numerical investigation of fluid flow and heat transfer around a solid circular cylinder utilizing nanofluid. *Int Commun Heat Mass Transfer*. 2011;38(9):1296-1304. doi:10.1016/j.icheatmasstransfer.2011.06.007
- Kasaeian A, Daneshzarian R, Mahian O, et al. Nanofluid flow and heat transfer in porous media: a review of the latest developments. *Int J Heat Mass Transf*. 2017;107:778-791. doi:10.1016/j.ijheatmasstransfer.2016.11.074
- Bao L, Zhong C, Jie P, Hou Y. The effect of nanoparticle size and nanoparticle aggregation on the flow characteristics of nanofluids by molecular dynamics simulation. *Adv Mech Eng*. 2019;11(11):1687814019889486. doi:10.1177/1687814019889486
- Basu AJ, Khalili A. Computation of flow through a fluid-sediment interface in a benthic chamber. *Phys Fluids*. 1999; 11(6):1395-1405. doi:10.1063/1.870004
- Chikh S, Boumedien A, Bouhadek K, Lauriat G. Analysis of fluid flow and heat transfer in a channel with intermittent heated porous blocks. *Heat Mass Transfer*. 1998;33(5-6):405-413. doi:10.1007/s002310050208



16. Vafai K, Kim SJ. On the limitations of the Brinkman-Forchheimer-extended Darcy equation. *Int J Heat Fluid Flow*. 1995;16(1):11-15. doi:[10.1016/0142-727X\(94\)00002-T](https://doi.org/10.1016/0142-727X(94)00002-T)
17. Bird RB, Stewart WE, Lightfoot EN. *Transport phenomena*. Wiley; 2006.
18. Brinkman HC. On the permeability of media consisting of closely packed porous particles. *Flow Turbul Combust*. 1949;1(1):81-86. doi:[10.1007/bf02120318](https://doi.org/10.1007/bf02120318)
19. Kang C, Mirbod P. Porosity effects in laminar fluid flow near permeable surfaces. *Phys Rev E*. 2019;100(1):013109. doi:[10.1103/PhysRevE.100.013109](https://doi.org/10.1103/PhysRevE.100.013109)
20. Hall C, Mackowiak J. *Fluid dynamics of packed columns: principles of the fluid dynamic Design of Columns for gas/liquid and liquid/liquid systems*. Springer; 2009. doi:[10.1007/b98397](https://doi.org/10.1007/b98397)
21. Pak BC, Cho YI. Hydrodynamic and heat transfer study of dispersed fluids with submicron metallic oxide particles. *Exp Heat Transfer*. 1998;11(2):151-170. doi:[10.1080/08916159808946559](https://doi.org/10.1080/08916159808946559)
22. Masoumi N, Sohrabi N, Behzadmehr A. A new model for calculating the effective viscosity of nanofluids. *J Phys D Appl Phys*. 2009;42(5):055501. doi:[10.1088/0022-3727/42/5/055501](https://doi.org/10.1088/0022-3727/42/5/055501)
23. Maerefat M, Mahmoudi SY, Mazaheri K. Numerical simulation of forced convection enhancement in a pipe by porous inserts. *Heat Trans Eng*. 2010;32(1):45-56. doi:[10.1080/01457631003732854](https://doi.org/10.1080/01457631003732854)
24. Kaur J, Melnik R, Tiwari AK. Forced convection heat transfer study of a blunt-headed cylinder in non-Newtonian power-law fluids. *Int J Chem Reactor Eng*. 2021;19(7):673-688. doi:[10.1515/ijcre-2020-0170](https://doi.org/10.1515/ijcre-2020-0170)
25. Kaur J, Ratan JK, Tiwari AK. Heat transfer study from square cylinder immersed in water-based nanofluid: effect of orientation. *Chem Process Eng*. 2022;43:243-250.
26. Sasmal C. Effects of axis ratio, nanoparticle volume fraction and its size on the momentum and heat transfer phenomena from an elliptic cylinder in water-based CuO nanofluids. *Powder Technol*. 2017;313:272-286. doi:[10.1016/j.powtec.2017.02.063](https://doi.org/10.1016/j.powtec.2017.02.063)
27. Sasmal C. Analysis of the effects of inclination angle, nanoparticle volume fraction and its size on forced convection from an inclined elliptic cylinder in aqueous nanofluids. *J Thermal Anal Calorimet*. 2018;136(3):1433-1445. doi:[10.1007/s10973-018-7750-5](https://doi.org/10.1007/s10973-018-7750-5)
28. Kaur R, Chandra A, Sharma S. Momentum transfer across a semi-circular porous cylinder attached to a channel wall. *Mechanica*. 2021;56(9):2219-2241. doi:[10.1007/s11012-021-01369-5](https://doi.org/10.1007/s11012-021-01369-5)
29. Anirudh K, Dhinakaran S. Effects of Prandtl number on the forced convection heat transfer from a porous square cylinder. *Int J Heat Mass Transf*. 2018;126:1358-1375. doi:[10.1016/j.ijheatmasstransfer.2018.06.003](https://doi.org/10.1016/j.ijheatmasstransfer.2018.06.003)
30. Anirudh K, Dhinakaran S. On the onset of vortex shedding past a two-dimensional porous square cylinder. *J Wind Eng Ind Aerodyn*. 2018;179:200-214. doi:[10.1016/j.jweia.2018.03.004](https://doi.org/10.1016/j.jweia.2018.03.004)
31. Sasmal C, Nirmalkar N. Momentum and heat transfer characteristics from heated spheroids in water based nanofluids. *Int J Heat Mass Transf*. 2016;96:582-601. doi:[10.1016/j.ijheatmasstransfer.2016.01.054](https://doi.org/10.1016/j.ijheatmasstransfer.2016.01.054)
32. Koteswara Rao P, Sasmal C, Sahu AK, Chhabra RP, Eswaran V. Effect of power-law fluid behavior on momentum and heat transfer characteristics of an inclined square cylinder in steady flow regime. *Int J Heat Mass Transf*. 2011;54(13-14):2854-2867. doi:[10.1016/j.ijheatmasstransfer.2011.03.013](https://doi.org/10.1016/j.ijheatmasstransfer.2011.03.013)

How to cite this article: Kaur J, Ratan JK, Melnik R, Tiwari AK. Hydrodynamic analysis of nanofluids flow over 45° inclined porous square cylinder using Darcy–Brinkman–Forchheimer model. *Asia-Pac J Chem Eng*. 2024;19(3):e3042. doi:[10.1002/apj.3042](https://doi.org/10.1002/apj.3042)